



MASTER THESIS · 2026

PDPA Ontology Design

A knowledge model for GraphRAG question answering on the Personal Data Protection Act



Presented by: [Your name]

Purpose of the ontology

Why the ontology exists, and what it must be able to answer



1

Knowledge base for GraphRAG

It stores PDPA concepts as a connected graph.

2

From text to structure

It turns the law text into classes, relations, and instances.

3

Traceable answers

Every term links to its exact law section, so the system can cite the law.

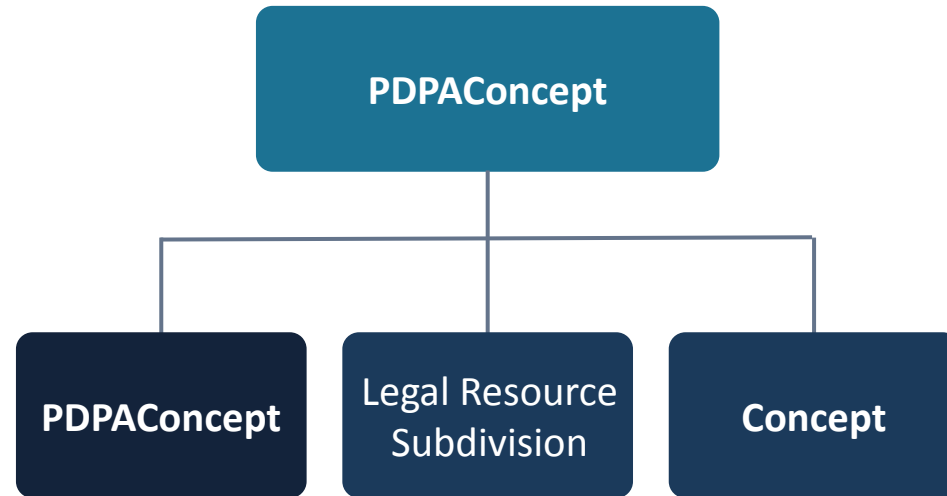
Competency questions

The design is complete if it can answer these: sample question

- Which section governs cross-border data transfer?
- What consent is needed for sensitive data?
- Which rights can a data subject exercise?
- When does the PDPA not apply (exemptions)?

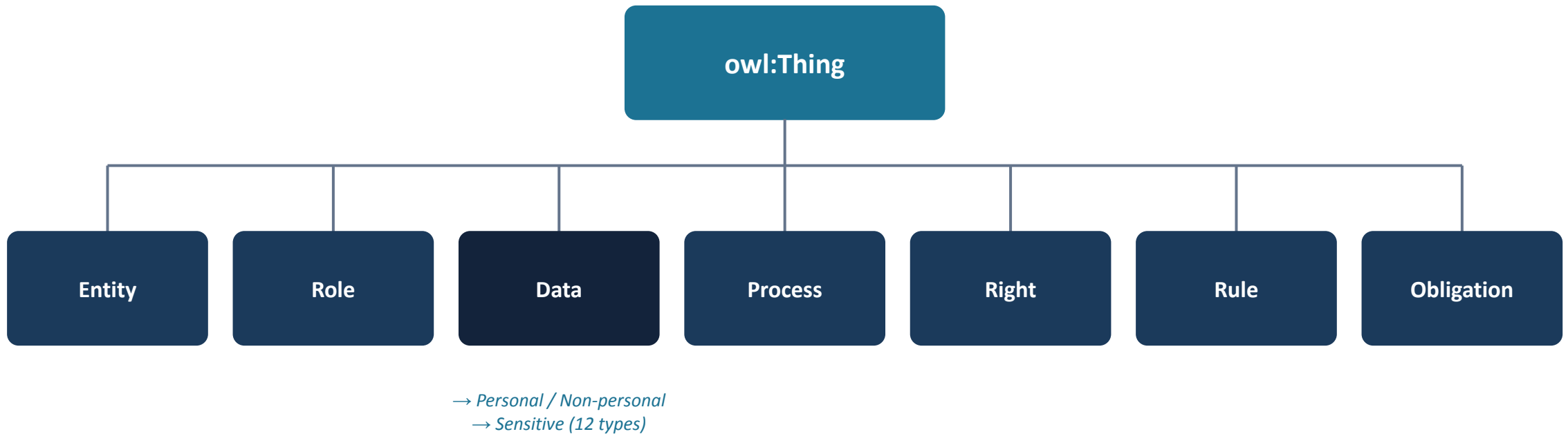
Ontology architecture

Three roots, with seven core branches under PDPAConcept



Ontology architecture

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Other roots: Concept · LegalResourceSubdivision (*Part / Chapter / Section — the law's own structure*)

The ontology at a glance

Size and coverage show a real, non-trivial model

85

Classes

35

Object properties

77

Individuals (37 sections)

1,184

Triples / axioms

18

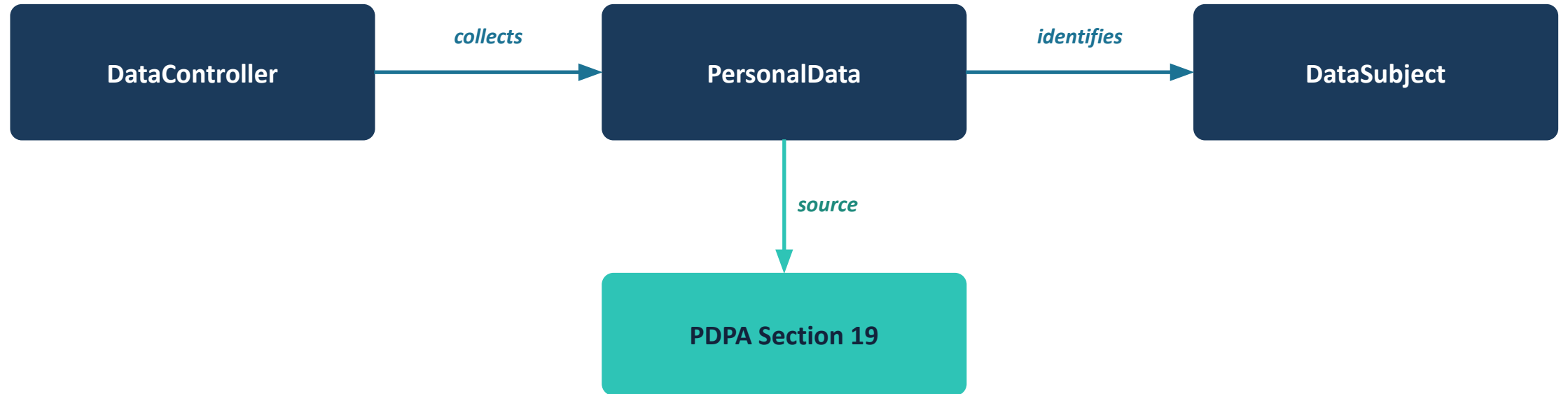
OWL restrictions

4

Standards reused

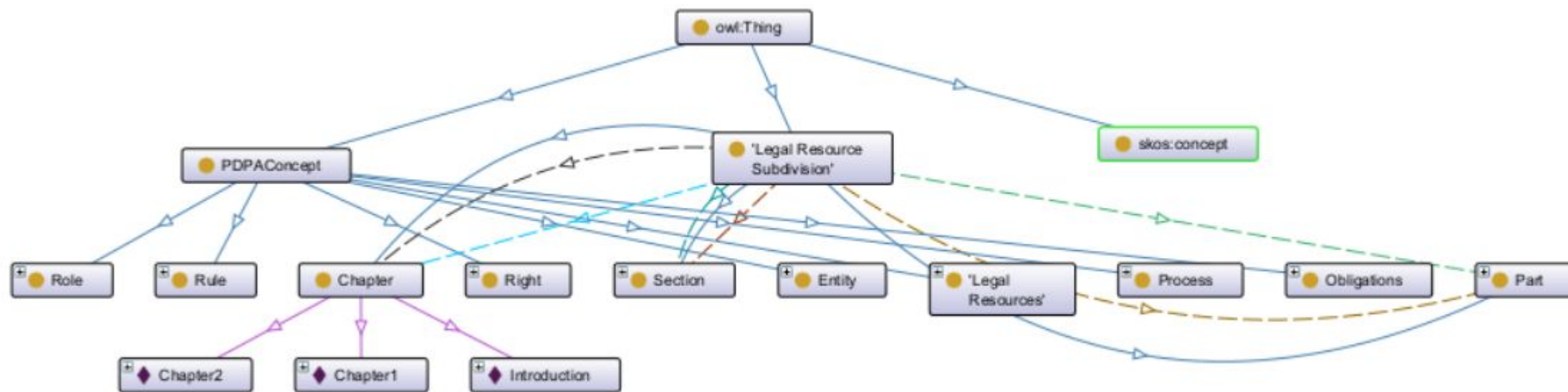
One example, end to end

How concepts, relations, and the law source connect



Legal traceability. Every term and relation links to its source section. So an answer can cite the exact law — the key strength of the design.

Full ontology graph



Why the design is sound

Three simple checks prove the ontology is built properly

1

Reuses standards

Built on OWL and RDFS. Reuses SKOS (skos:definition) and ELI (eli:title) for legal metadata. Not an isolated, custom model.

2

Logically consistent

Runs under an OWL reasoner (HermiT / Pellet) with no unsatisfiable classes. Add the reasoner screenshot as evidence.

3

Answers the questions

SPARQL queries return the correct sections for each competency question. The ontology does its job.

Tip: run the reasoner in Protégé and keep the screenshot — it is strong proof for your defence.



Key message

The ontology is not just a vocabulary.

Every concept links back to its law section — so the system can cite the law.

Thank you · Questions welcome

